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1. Experiment Title

Use Function Types, Scope, Closures, Try-Catch Exception Handling, and Build a Palindrome Checker

2. Software / Tools Required

Visual Studio Code

HTML5

CSS3

JavaScript

Google Chrome

3. Experiment Program Code

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Palindrome Checker | Glassmorphism</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <div class="background-mesh">
    <div class="orb orb-1"></div>
    <div class="orb orb-2"></div>
  </div>

  <div class="glass-card">
    <div class="card-header">
      <h1>Palindrome Checker</h1>
      <p id="heading"></p>
    </div>

    <div class="input-group">
      <input
        type="text"
        id="word"
        placeholder="Enter a word (e.g., racecar)"
        autocomplete="off"
        aria-label="Word to check">

      <button onclick="checkPalindrome()" class="glass-btn">
        <span>Check</span>
        <div class="btn-shine"></div>
      </button>
    </div>

    <div id="result" class="result-area" aria-live="polite"></div>
  </div>

  <script src="script.js"></script>
</body>
</html>
```

script.js

```
function reverseString(str) {  
    return str.split("").reverse().join("");  
}
```

```
function message() {  
    const msg = "Palindrome Checker";  
    return () => msg;  
}
```

// Initialize heading using closure

```
const showMessage = message();
```

```
document.getElementById("heading").innerText = showMessage();
```

```
function checkPalindrome() {  
    const wordInput = document.getElementById("word");  
    const resultEl = document.getElementById("result");  
    const word = wordInput.value.trim();
```

// Reset previous state

```
resultEl.className = "result-area";
```

```
resultEl.innerText = "";
```

```
try {
```

```
    if (!word) {
```

```
        throw new Error("Please enter a word.");
```

```
    }
```

```
    const reversed = reverseString(word);
```

```
    const isPalindrome = word.toLowerCase() === reversed.toLowerCase();
```

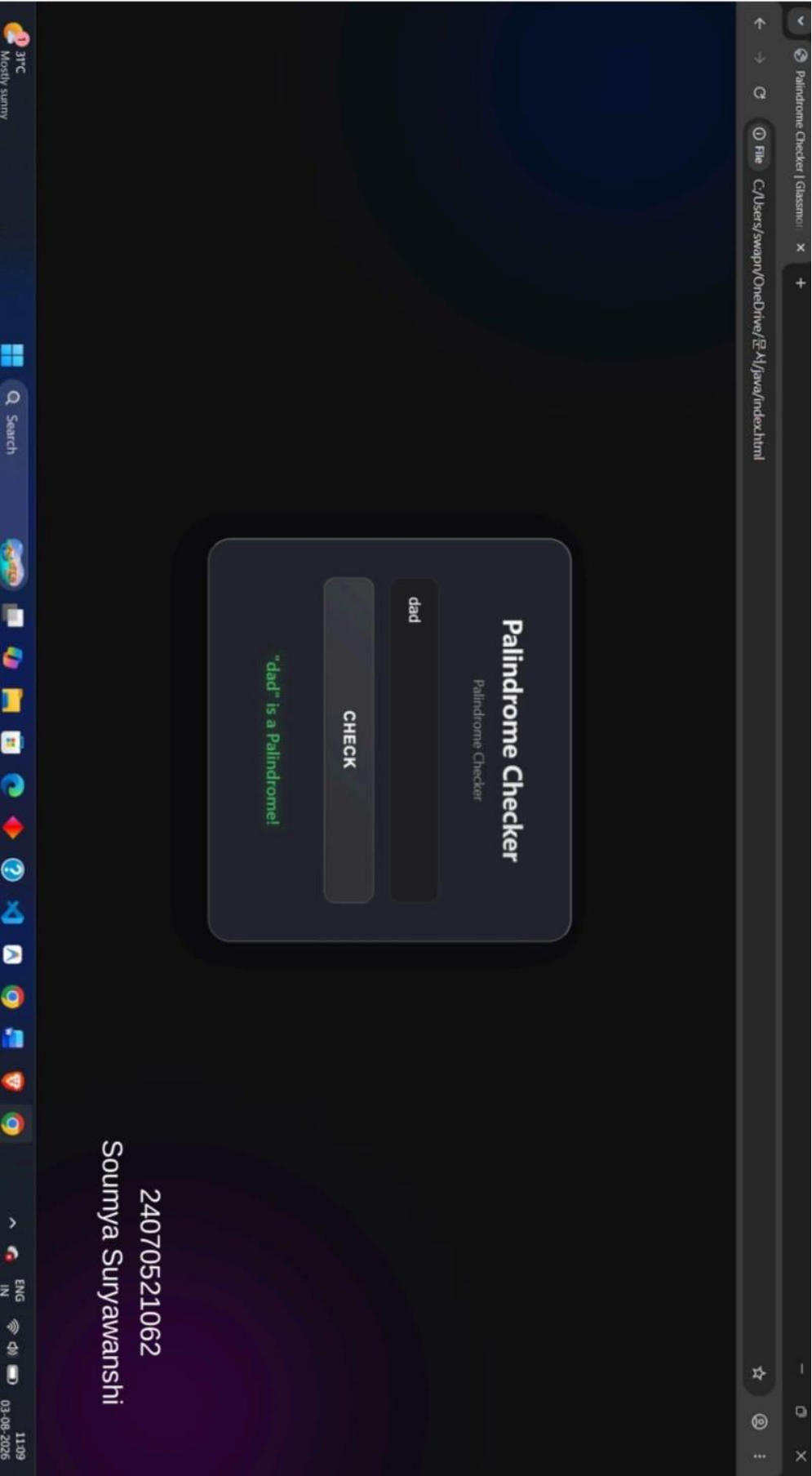
```
    if (isPalindrome) {
        showResult(`${word}" is a Palindrome!`, "success");
    } else {
        showResult(`${word}" is Not a Palindrome.`, "error");
    }

} catch (error) {
    showResult(error.message, "info");
    wordInput.focus();
}
}

function showResult(text, type) {
    const resultEl = document.getElementById("result");
    resultEl.innerText = text;
    resultEl.className = `result-area ${type} visible`;
}

// Allow Enter key to trigger palindrome check
document.getElementById("word").addEventListener("keypress", function (event) {
    if (event.key === "Enter") {
        checkPalindrome();
    }
});
```

4. Output



5. Case Study Title

Vehicle number Validation checker

6. Case Study Program Cod

[index.html](#)

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Sarathi Parivahan - Vehicle Registration Validation</title>

  <link rel="stylesheet" href="style.css">

  <link href="https://fonts.googleapis.com/css2?family=Roboto:wght@400;500;700&display=swap"
rel="stylesheet">
</head>

<body>

<div class="govt-container">

  <header class="govt-header">

    <div class="emblem-section">
      
      </div>

      <div class="header-text">
        <h1>Ministry of Road Transport and Highways</h1>
        <h2>Department of Road Transport</h2>
        <h3>Sarathi Parivahan Sewa</h3>

        <p class="subtitle">
          Vehicle Registration Number Validation Service
        </p>
      </div>

    </header>

    <div class="content-wrapper">
```

```
<div class="form-panel">
```

```
<div class="panel-header">
```

```
<h4>Online Vehicle Registration Validation</h4>
```

```
</div>
```

```
<div class="form-body">
```

```
<p class="instruction-text">
```

```
  Please enter the Vehicle Registration Number to validate  
  its format against Indian standards.
```

```
</p>
```

```
<div class="input-group">
```

```
<label
```

```
  for="registrationInput"
```

```
  class="form-label">
```

```
  Vehicle Registration Number
```

```
</label>
```

```
<div class="input-control-wrapper">
```

```
<input
```

```
  type="text"
```

```
  id="registrationInput"
```

```
  placeholder="e.g. MH31AB1234"
```

```
  maxlength="10"
```

```
  class="form-input">
```

```
<button
```

```
  id="validateBtn"
```

```
  class="govt-btn">
```

```
  Validate Now
```

```
</button>
```

```
</div>
```

```
</div>
```

```
<div
  id="result"
  class="result-box hidden"
  role="alert">
```

```
</div>
```

```
<div class="info-box">
```

```
  <h5>Validation Criteria:</h5>
```

```
  <ul class="criteria-list">
```

```
    <li>
```

```
      Format:
```

```
      <strong>XX 00 XX 0000</strong>
```

```
      (State-District-Series-Number)
```

```
    </li>
```

```
    <li>
```

```
      First 2 characters:
```

```
      State Code (Alphabets)
```

```
    </li>
```

```
    <li>
```

```
      Next 2 characters:
```

```
      District Code (Digits)
```

```
    </li>
```

```
    <li>
```

```
      Next 2 characters:
```

```
      Series Code (Alphabets)
```

```
    </li>
```

```
    <li>
```

```
      Last 4 characters:
```

```
      Unique Vehicle Number (Digits)
```

```
    </li>
```

```
  </ul>
```

```
</div>
```

```
</div>
```

</div>

</div>

<footer class="govt-footer">

<p>

© 2026 Ministry of Road Transport and Highways,
Govt of India.

</p>

<p class="footer-note">

System generated validation for informational purposes only.

</p>

</footer>

</div>

<script src="script.js"></script>

</body>

</html>

script.js

```
document.addEventListener('DOMContentLoaded', () => {
```

```
  const inputField = document.getElementById('registrationInput');  
  const validateBtn = document.getElementById('validateBtn');  
  const resultDiv = document.getElementById('result');
```

```
  validateBtn.addEventListener('click', () => {
```

```
    const rawInput = inputField.value;
```

```
    // Reset previous result  
    resultDiv.className = 'result-box';
```

```
    try {
```

```
      // Check if input is empty  
      if (!rawInput || rawInput.trim() === '') {  
        throw new Error('Registration number cannot be empty');  
      }
```

```
      // Convert to uppercase and remove extra spaces
```



```

const registrationNumber = rawInput.trim().toUpperCase();

// Validate total length
if (registrationNumber.length !== 10) {
    throw new Error('Length must be exactly 10 characters');
}

// Split into components
const stateCode = registrationNumber.substring(0, 2);
const districtCode = registrationNumber.substring(2, 4);
const series = registrationNumber.substring(4, 6);
const vehicleNumber = registrationNumber.substring(6, 10);

// State Code Validation
if (!/^[A-Z]{2}$/.test(stateCode)) {
    throw new Error('First 2 characters must be uppercase alphabets');
}

// District Code Validation
if (!/^[0-9]{2}$/.test(districtCode)) {
    throw new Error('Next 2 characters must be digits');
}

// Series Validation
if (!/^[A-Z]{2}$/.test(series)) {
    throw new Error('Next 2 characters must be uppercase alphabets');
}

// Vehicle Number Validation
if (!/^[0-9]{4}$/.test(vehicleNumber)) {
    throw new Error('Last 4 characters must be digits');
}

// Success Message
resultDiv.classList.add('valid');

resultDiv.innerHTML =
    '<strong>Valid!</strong><br>' +
    registrationNumber;
}

catch (error) {

    resultDiv.classList.add('invalid');

    resultDiv.innerHTML =
        '<strong>Invalid!</strong><br>' +
        error.message;
}

```

```

});

// Allow Enter key to validate
inputField.addEventListener('keypress', (event) => {

    if (event.key === 'Enter') {
        validateBtn.click();
    }

});

});

```

7. Output

The screenshot shows a web browser window displaying the Sarathi Parivahan - Vehicle Registration Number Validation Service. The page has a dark blue header with the Ministry of Road Transport and Highways logo and text. Below the header, there is a section titled "ONLINE VEHICLE REGISTRATION VALIDATION". It contains a text input field with the value "MH29AB3942" and a "VALIDATE NOW" button. Below the input field, a green box displays "Valid!" and "MH29AB3942". At the bottom, there is a "VALIDATION CRITERIA" section with a list of rules: Format: XX 00 XX 0000 (State-District-Series-Number), First 2 characters: State Code (Alphabets), Next 2 characters: District Code (Digits), Next 2 characters: Series Code (Alphabets), and Last 4 characters: Unique Vehicle Number (Digits). The footer of the page mentions "© 2026 Ministry of Road Transport and Highways, Govt of India" and "System generated validation for informational purposes only."

8. Result / Conclusion

The experiment and case study were implemented successfully. Functions, closures, scope, exception handling and DOM manipulation were demonstrated, and the vehicle registration validation worked as expected.